EEEM073 - AI And Sustainability

Course Work: PM 2.5 Concentration Prediction

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1. Introduction

1.1 Problem Statement and Application

Air pollution is one of the most critical environmental and public health challenges of the 21st century. Fine particulate matter (PM2.5), defined as particles with a diameter of 2.5 micrometres or less, is considered the most harmful air pollutant due to its ability to penetrate deep into the lungs and bloodstream, leading to respiratory illness, cardiovascular disease, and premature mortality. According to World Health Organization guidelines, annual PM2.5 levels should not exceed 5 µg/m³; however, cities such as Lahore frequently experience daily concentrations exceeding 200-400 µg/m³ during peak smog periods [3][5]. Additionally, crop residue burning is a major contributor to air pollution and seasonal smog formation in the Indo-Gangetic region, significantly increasing PM2.5 concentrations during winter months [6].
This project develops an AI-based time-series forecasting and decision support system to predict next-day PM2.5 levels in Lahore. By forecasting pollution levels in advance, the system enables proactive interventions such as public health advisories, traffic restrictions, and industrial regulation, which are significantly more effective when implemented before exposure occurs.

Problem Definition

The problem is formulated as a supervised regression and time-series forecasting task, with the objective:

To predict next-day PM2.5 concentrations in Lahore using historical environmental and fire activity data, and to generate actionable decision-support insights.

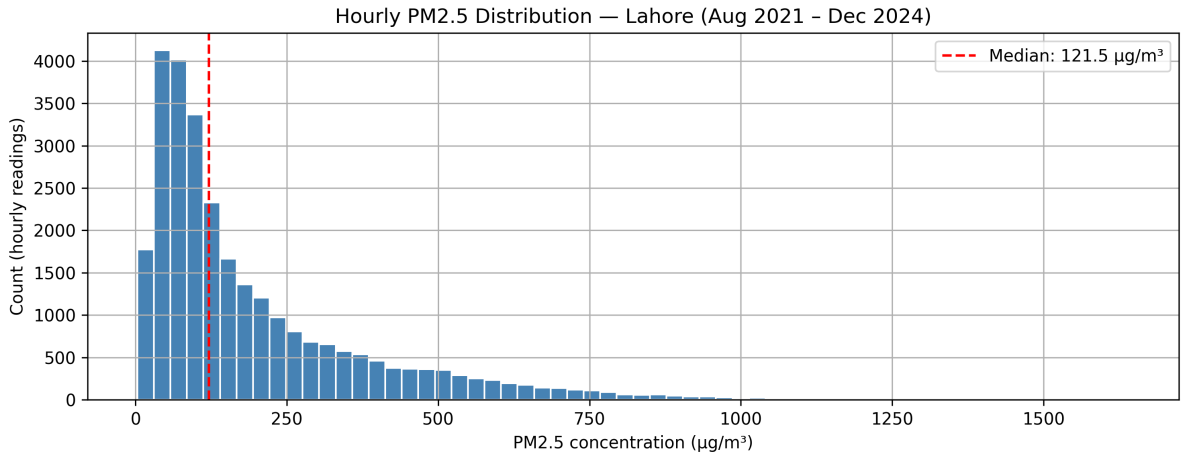


Figure 1 Shows hourly PM2.5 Concentration over the years 2021 - 2024 in Lahore, Pakistan

Objectives and Methodology

The primary objective of this project is to design and implement an AI-driven time-series decision support system for PM_2.5 forecasting in Lahore. The methodology is structured into the following sequential stages:

- **Data Preparation & Integration**: Standardising multi-source datasets (air quality, meteorology, and NASA MODIS fire activity) through rigorous cleaning, outlier treatment, and temporal aggregation into daily intervals.
- **Feature Engineering**: Enhancing predictive signals by introducing temporal lags, rolling averages, and spatially-filtered fire metrics to capture environmental dependencies.

- **Exploratory Data Analysis**: Analysing seasonal trends and correlations between PM_2.5, meteorological variables, and regional fire activity to validate dataset integrity.
- **Model Development & Tuning**: Implementing and benchmarking diverse architectures, including Linear Regression, LSTM, CNN1D, Transformer, and GRU models, refined through hyperparameter and sequence window optimisation.
- **Evaluation & XAI Analysis**: Validating performance using MAE, RMSE, and R^2 metrics alongside visual residual analysis. Interpretability is established via SHAP and LIME to identify the key drivers of pollution events.
- **Sustainable AI & Decision Support**: Deploying model compression techniques to reduce computational overhead, ultimately translating model outputs into actionable public health and policy recommendations.

Research Questions

1. How accurately can PM2.5 levels be predicted using environmental and fire data?
2. What are the most influential factors driving pollution in Lahore?
3. To what extent does regional fire activity impact urban air quality?
4. How can predictions be translated into actionable interventions?

1.2 Alignment with UN Sustainable Development Goals

This project contributes to:

- SDG 3: Good Health and Well-being - enabling early warnings for pollution exposure
- SDG 11: Sustainable Cities and Communities - supporting urban air quality management
- SDG 13: Climate Action - addressing emissions linked to biomass burning

The project demonstrates how AI can transition environmental management from reactive to proactive decision-making, aligning strongly with sustainability objectives [4].

1.3 Datasets and Data Processing

Air Quality and Weather Dataset

- Source: Kaggle [1]
- Time period: 2021-2024
- Size: ~28,000 hourly observations
- Features:
    - Pollutants: PM2.5 (target), PM10, NO_2, SO_2, CO, O_3
    - Weather: temperature, humidity, wind speed, pressure

This dataset captures local environmental conditions and forms the core input for prediction.

Fire Activity Dataset

- Source: NASA FIRMS [2]
- Data type: Satellite-detected fire events (MODIS & VIIRS)
- Time period: 2021-2024 (downloaded yearly)
- Location: Pakistan

To construct a usable fire dataset aligned with the project objectives, the following steps were performed:

1. **Data Collection**
   Fire data was downloaded separately for each year (2021-2024) to ensure full temporal coverage.
2. **Concatenation**
   All yearly datasets were merged into a single dataset to create a continuous time-series of fire activity.
3. **Spatial Filtering**
   The dataset was filtered using latitude and longitude constraints to retain only fire events within Lahore and its surrounding regions.
   This step ensures:
   - relevance to local air quality
   - inclusion of nearby agricultural burning activity
4. **Temporal Aggregation**
   Since the air quality dataset is aggregated at a daily level for modelling, the fire dataset was converted from event-based observations to daily features, including:
   - total daily fire count
   - average fire intensity (FRP)
   - maximum fire intensity
   - high-confidence fire detections
5. **Alignment with Air Quality Data**
   The processed daily fire dataset was then aligned temporally with the air quality and weather dataset to enable merging.

1.4 Dataset Complexity and Justification

The final dataset is a multi-source, multi-dimensional time-series dataset, with a total of 21 features combining:

- Local environmental measurements
- Meteorological conditions
- Satellite-derived regional fire activity

Complexity Characteristics

- Temporal complexity: hourly → daily aggregation
- Spatial complexity: filtering based on geographic constraints
- Feature complexity: engineered features (lags, rolling statistics, fire metrics)
- Data integration: merging heterogeneous datasets

This level of complexity reflects the multi-causal nature of air pollution. Air quality data alone captures symptoms of pollution, fire data introduces external causal signals, and weather data explains dispersion dynamics. Together, these datasets allow the model to learn meaningful relationships rather than simple correlations.

1.5 Limitations and Critical Evaluation

Despite its strengths, the dataset has limitations:

- Fire data does not explicitly label crop burning, requiring interpretation
- Cross-border pollution sources may not be fully captured
- Traffic and industrial emissions are not explicitly included

However, these limitations are acknowledged and do not prevent the dataset from effectively addressing the research objectives. Instead, they highlight opportunities for future work.

2. Data Understanding, Pre-processing and Exploratory Analysis

2.1 Dataset Loading and Initial Inspection

The primary dataset was loaded as a time-indexed data frame using appropriate parsing parameters to ensure correct datetime formatting and encoding consistency. The dataset consists of 28,268 hourly observations spanning August 2021 to December 2024, with 18 variables: eight pollutant variables (CO, NO, NO_2, ozone, SO_2, PM2.5, PM10, ammonia) and eight meteorological variables (temperature, humidity, dew point, precipitation, surface pressure, wind speed, wind direction, shortwave solar radiation).

Column	Type	Unit	Description
datetime	datetime	—	Hourly timestamp of the observation (DD/MM/YYYY HH:MM, UTC)
main.aqi	int	1–5	Air Quality Index category: 1=Good, 2=Fair, 3=Moderate, 4=Poor, 5=Very Poor
components.co	float	µg/m³	Carbon monoxide concentration — primarily from vehicle exhaust and combustion
components.no	float	µg/m³	Nitric oxide concentration — a precursor to NO_2, emitted from combustion sources
components.no2	float	µg/m³	Nitrogen dioxide concentration — traffic and industrial emissions indicator
components.o3	float	µg/m³	Ozone concentration — secondary pollutant formed by photochemical reactions
components.so2	float	µg/m³	Sulphur dioxide concentration — linked to industrial activity and fuel burning
components.pm2_5	float	µg/m³	Fine particulate matter (diameter < 2.5 µm) — penetrates deep into the lungs
components.pm10	float	µg/m³	Coarse particulate matter (diameter < 10 µm) — includes dust, pollen, smoke
components.nh3	float	µg/m³	Ammonia concentration — primarily from agricultural activities
temperature_2m	float	°C	Air temperature measured at 2 m above ground
relative_humidity_2m	float	%	Relative humidity at 2 m — affects pollutant dispersion and formation
dew_point_2m	float	°C	Dew point temperature at 2 m — indicates moisture content of the atmosphere
precipitation	float	mm	Total precipitation — rain/snow can washout particulates from the atmosphere
surface_pressure	float	hPa	Atmospheric pressure at the surface — low pressure promotes pollutant build-up
wind_speed_10m	float	m/s	Wind speed at 10 m — higher wind speed disperses pollutants
wind_direction_10m	float	°	Wind direction at 10 m (0–360°, where 0/360=North, 90=East)
shortwave_radiation	float	W/m²	Solar shortwave radiation — drives photochemical ozone formation

Figure 2 Shows attributes in the Primary Air Quality and Weather dataset

In parallel, four annual MODIS fire datasets (2021-2024) were loaded and concatenated into a unified dataset comprising 47,353 records. The acquisition date field was converted to datetime format, and a year identifier was retained to preserve temporal traceability.

Column	Type	Unit	Description
latitude	float	decimal °	Latitude of the fire pixel centroid
longitude	float	decimal °	Longitude of the fire pixel centroid
brightness	float	Kelvin	Channel 21/22 brightness temperature — proxy for fire temperature
scan	float	km	Along-scan spatial resolution of the fire pixel
track	float	km	Along-track spatial resolution of the fire pixel
acq_date	date	YYYY-MM-DD	Date the fire was detected by the satellite
acq_time	int	HHMM (UTC)	Time of satellite overpass when the fire was detected
satellite	str	—	Satellite platform: Terra (morning overpass) or Aqua (afternoon overpass)
instrument	str	—	Sensor name — always MODIS for this dataset
confidence	int	%	Detection confidence: < 30 = low, 30–79 = nominal, ≥ 80 = high
version	float	—	FIRMS collection and processing version number
bright_t31	float	Kelvin	Channel 31 brightness temperature — represents background/ambient surface temperature
frp	float	MW	Fire Radiative Power — measures the rate of radiant energy released; proxy for fire intensity
daynight	str	D / N	Whether detection occurred during daytime (D) or night-time (N)
type	int	0–3	Fire type: 0=presumed vegetation fire, 1=active volcano, 2=static land source, 3=offshore
year	int	—	Year of detection — added during data loading to identify source file

Figure 3 Shows attributes in the NASA FIRMS MODIS Dataset

An initial inspection was performed to assess:

- Data types and feature structure
- Missing values and null distribution
- Value ranges for physical plausibility

This step provided a comprehensive understanding of data quality and ensured that subsequent preprocessing decisions were grounded in the observed characteristics of the dataset.

2.2 Temporal Aggregation: Hourly to Daily Resolution

To ensure compatibility between datasets and reduce high-frequency noise, the hourly air quality data was aggregated to daily resolution. Aggregation rules were selected based on physical interpretation:

- Precipitation: summed to capture total daily rainfall
- All other continuous variables: averaged to represent daily conditions

This transformation resulted in 1,196 daily observations and enabled direct integration with daily fire activity data. Daily aggregation balances:

- noise reduction
- preservation of temporal trends
- compatibility with forecasting objectives

A limitation is the use of arithmetic mean for wind direction instead of circular averaging, which is acknowledged but does not significantly affect downstream modelling.

2.3 MODIS Fire Data Integration

The MODIS fire dataset was processed to extract regionally relevant and high-confidence fire activity.

1. Spatial Filtering: A Haversine distance function was used to retain fire detections within a 300 km radius of Lahore. This captures the Punjab agricultural burning belt while excluding distant, irrelevant fires.
2. Quality Filtering: Additionally, only detections with a satellite confidence score ≥ 30% and fire_type = 0 (vegetation fires only, excluding industrial/gas flares) were retained, ensuring that only genuine agricultural burning events were included.
3. Temporal Aggregation: The filtered MODIS records were grouped by acquisition date and aggregated to five daily fire features: ‘fire_count’ (number of detections), ‘frp_sum’ (total Fire Radiative Power in MW), ‘frp_mean’, ‘frp_max’, and ‘fire_brightness_avg’ (mean MODIS brightness temperature). Days with no detections were assigned zero values, representing true absence of fire activity rather than missing data.
4. Integration: The fire dataset was merged with the daily air quality dataset via a left join, ensuring full temporal coverage.

2.4 Data Cleaning

2.4.1 Missing Data Handling

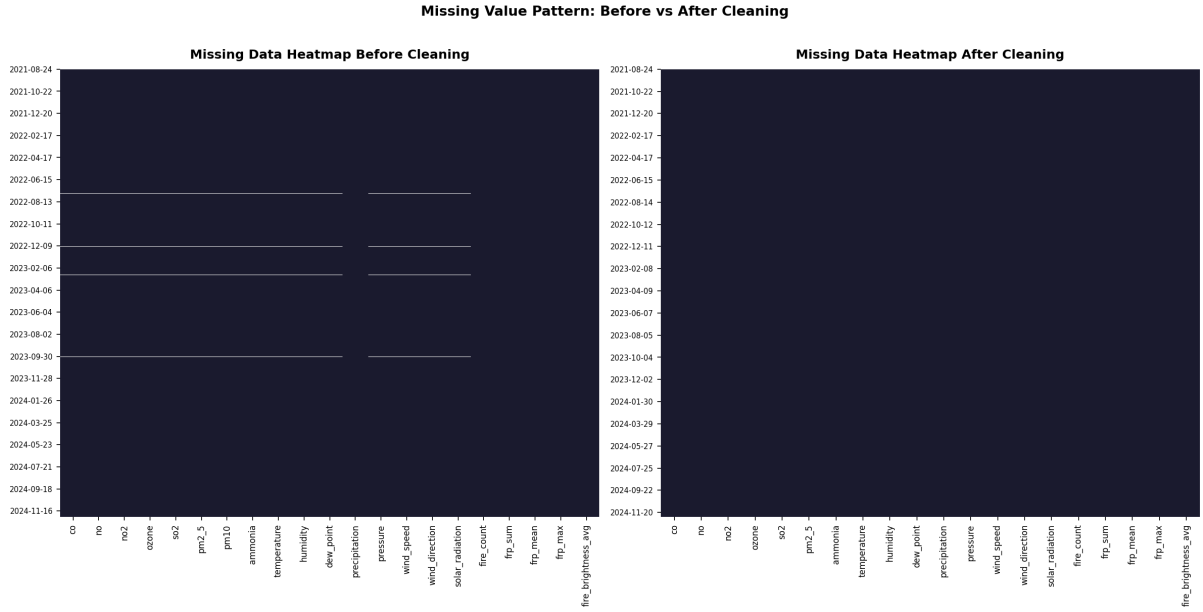


Figure 4 Show missing data heatmaps before and after cleaning

Following the merge and renaming, a missing data heatmap was generated to visualise the spatial and temporal distribution of null values across all 21 columns. The heatmap revealed 60 missing values across the dataset, precisely 4 missing values in each of 15 non-fire columns. This pattern confirms complete-row sensor outages rather than random column level gaps, consistent with brief API or monitoring station downtime. These 4 rows were removed as the first cleaning step, reducing the dataset from 1,196 to 1,192 rows.
Dropping was preferred over forward fill or interpolation because the missing rows represent complete-day outages where any imputed value would be a fabrication with no observational basis. Moreover, 0.33%, data loss is negligible and imputation in time series can introduce bias by smoothing real variation, particularly problematic for the seasonal PM2.5 spikes that are most important to model accurately. The after-cleaning heatmap confirmed zero remaining null values across all columns. A duplicate date check was also performed so that no duplicate date index entries were found.

2.4.2 Physical Validity Constraints

Domain-based constraints were applied:

- Pollutants >= 0: clipped to a lower bound of 0 μg/m³. Physically, atmospheric pollutant concentrations cannot be negative; negative values arise from sensor calibration errors.
- Humidity within [0,100]
- Extreme PM2.5 values exceeding 1,000 μg/m³ were capped at 1,000 μg/m³ as likely sensor malfunctions.
- Fire Columns >= 0

Unlike typical preprocessing pipelines, statistical outlier removal was intentionally avoided, as extreme values represent real smog events and are critical for model learning.

2.4.3 Feature Reduction

- Low variance features removed

- Highly correlated features (|r| > 0.95) eliminated: pm_10 column removed due to high correlation.

This improves model stability while preserving meaningful signals.

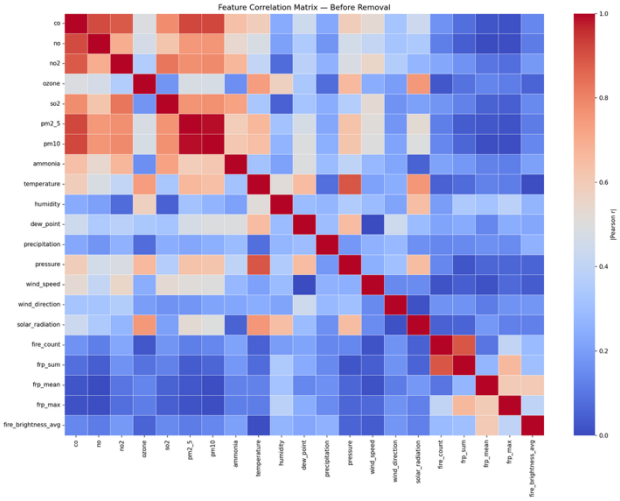


Figure 5 Feature Correlation Matrix before feature reduction

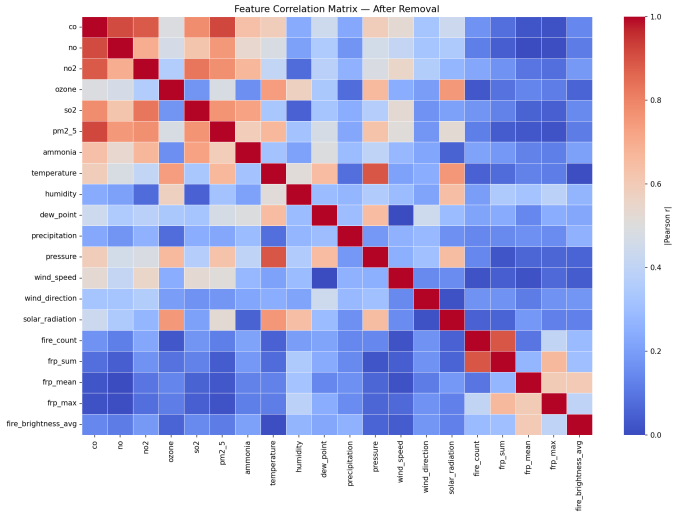


Figure 6 Feature Correlation matrix after feature reduction

2.5 Feature Engineering

Feature engineering was designed to capture temporal, seasonal, and external drivers of pollution contributing towards PM 2.5 concentration.

2.5.1 Temporal Features

Twelve temporal features were derived from the date index:

1. Calendar variables: year, month, day_of_week, day_of_year, week_of_year, is_weekend.
2. Seasonal binary flags: is_winter (December-February), is_spring (March-May), is_summer (June-August), is_autumn (September-November). These capture the four distinct pollution regimes driven by monsoon (summer, lowest PM2.5), post-monsoon (autumn, increasing), and winter temperature inversion (peak pollution).
3. Domain-specific flags: is_smog_season set to 1 for days in November, December, January, or on/after 15 October. This window reflects the combination of temperature inversions and reduced atmospheric mixing that characterises Lahore's annual smog crisis. is_crop_burning set to 1 for days between 15 October and 15 November, the peak period of post-harvest wheat and rice stubble burning in the Punjab agricultural belt. This flag provides the model with explicit awareness of the highest fire-emission period.

2.5.2 Cyclical Encoding

Sine/cosine transformations were applied to month, day of year, wind direction. This preserves cyclical relationships and improves model learning.

2.5.2 Lag Features

PM2.5 lag (1, 3, 7 days) and Fire activity lag features were added to capture temporal persistence and delayed effects. Providing the model with direct access to recent pollution history and fire activity before the look-back window of the sequence model begins. The lag-7 feature captures weekly cycles in human activity (e.g., industrial output, traffic) that contribute to PM2.5 variability. Lagging fire FRP by 1 day reflects the atmospheric transport time for smoke particles to reach Lahore from the Punjab burning belt.

2.5.3 Rolling Features

7-day rolling mean (short-term trend), 30-day rolling mean (seasonal baseline), rolling fire intensity all computed using shifted windows to prevent data leakage.

2.5.5 Final Adjustment

Initial rows affected by lag creation were removed, resulting in a clean dataset of 1,185 observations.

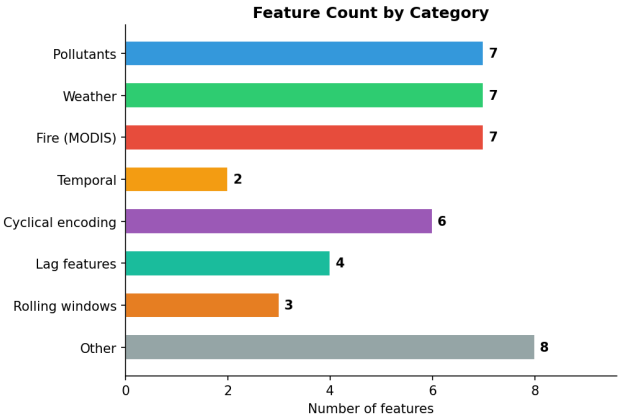


Figure 7 Total number of features by category after feature engineering

2.6 Exploratory Data Analysis

2.6.1 Target Variable Distribution

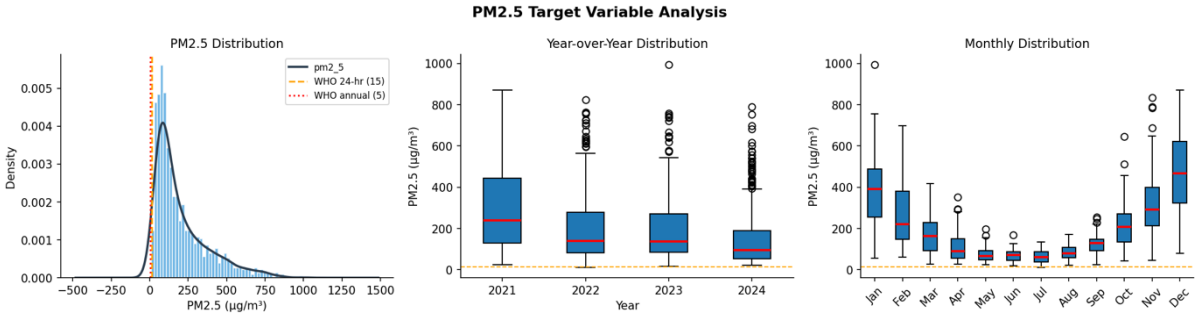


Figure 8 PM2.5 Distribution and Severity

The WHO 24-hour PM2.5 guideline of 15 µg/m³ was exceeded on 95.1% of all days in the dataset, and the annual mean guideline of 5 µg/m³ was always exceeded [3]. The maximum daily mean PM2.5 recorded was 994.88 µg/m³ illustrating the severity of Lahore's pollution crisis. The target distribution is right-skewed with a mean of 199.01 µg/m³ and a median of 134.29 µg/m³.

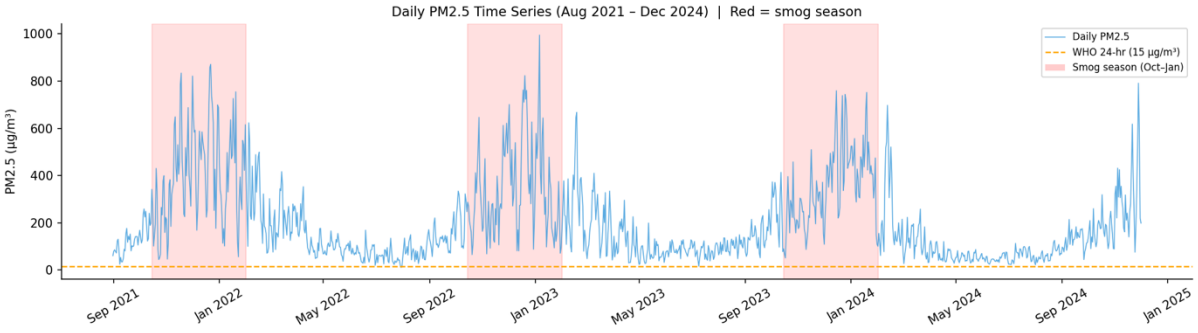


Figure 9 Shows Seasonality and Trends in PM 2.5 Concentration over the years

2.6.2 Temporal and Seasonal Patterns

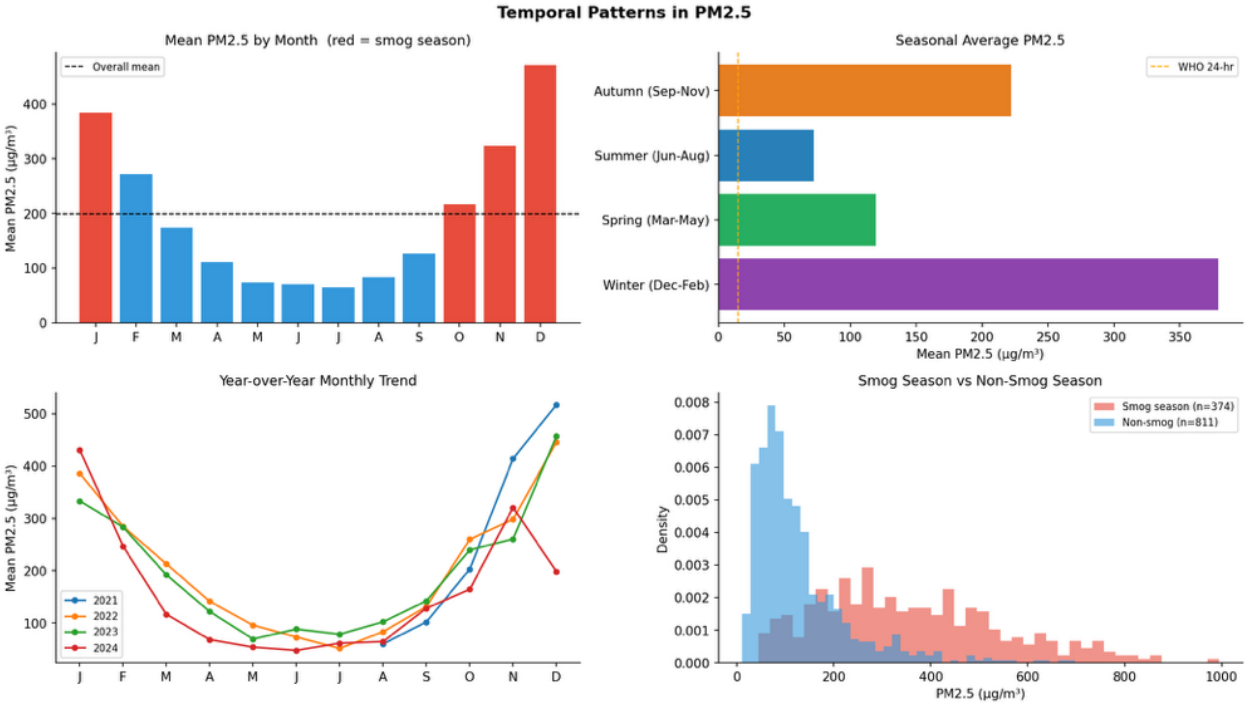


Figure 10 Seasonal patterns can be observed winter peaks (smog season) and summer lows (monsoon dispersion)

2.6.3 Fire Activity and PM 2.5 Relationship

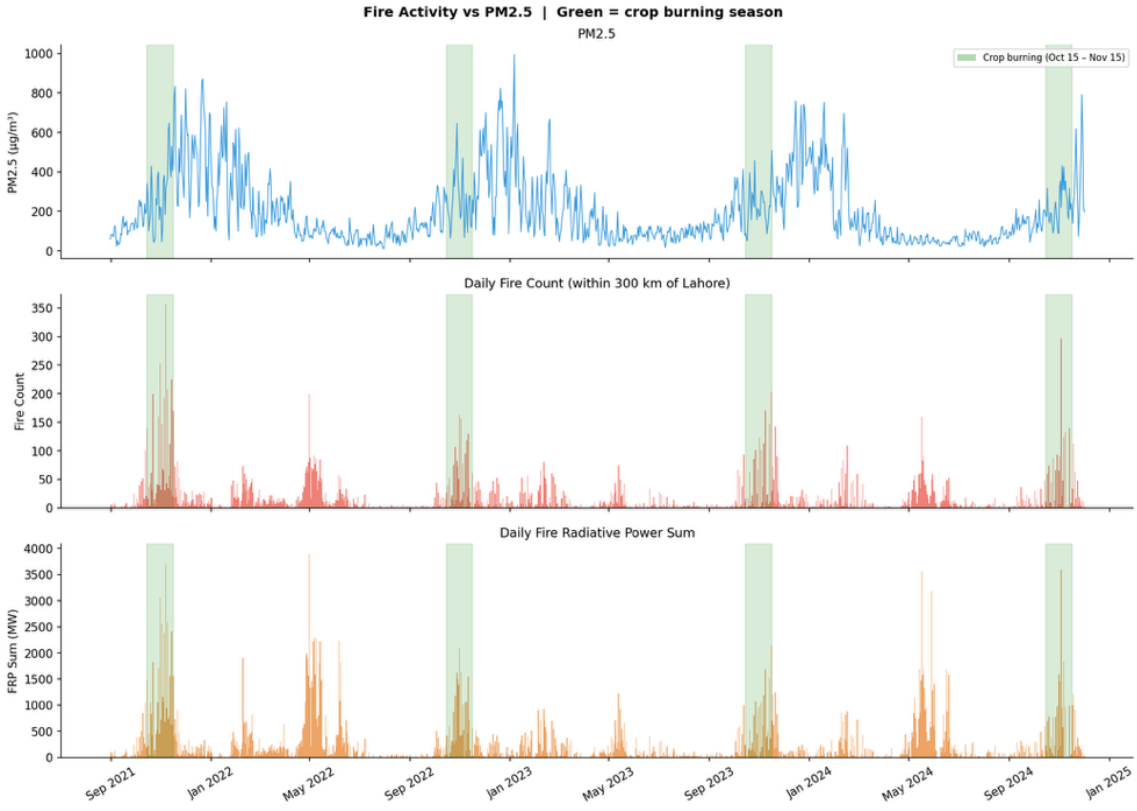


Figure 11 Fire activity shows strong temporal alignment with PM2.5 peaks, particularly during crop burning periods.

2.6.4 Meteorological Relationships

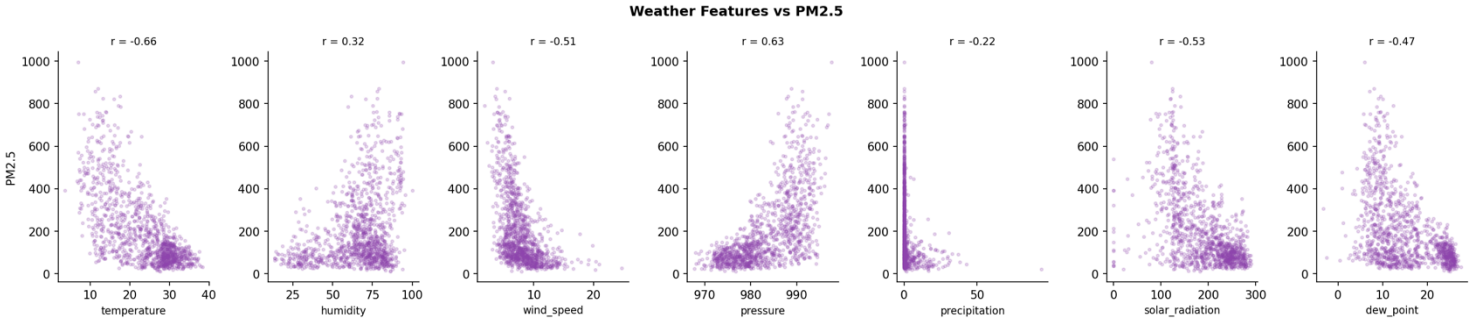


Figure 12 Scatter plots of meteorological variables versus daily PM 2.5

Temperature shows the strongest negative correlation. Low temperatures are associated with higher PM2.5, consistent with winter temperature inversions. Solar radiation also shows a negative relationship, as clear low-solar days in winter coincide with stagnant air masses. Humidity shows a positive correlation, consistent with hygroscopic growth of aerosol particles in humid conditions.

2.6.5 Autocorrelation Analysis

ACF and PACF plots show strong temporal dependence up to lag 7. Significant autocorrelation extends to approximately lag 7, with partial autocorrelations remaining significant at lags 1, 3, 6 and 7. This justifies the use of a 7-day look-back window for all sequence models. Model receives 7 days of historical features to predict the next day's PM2.5.

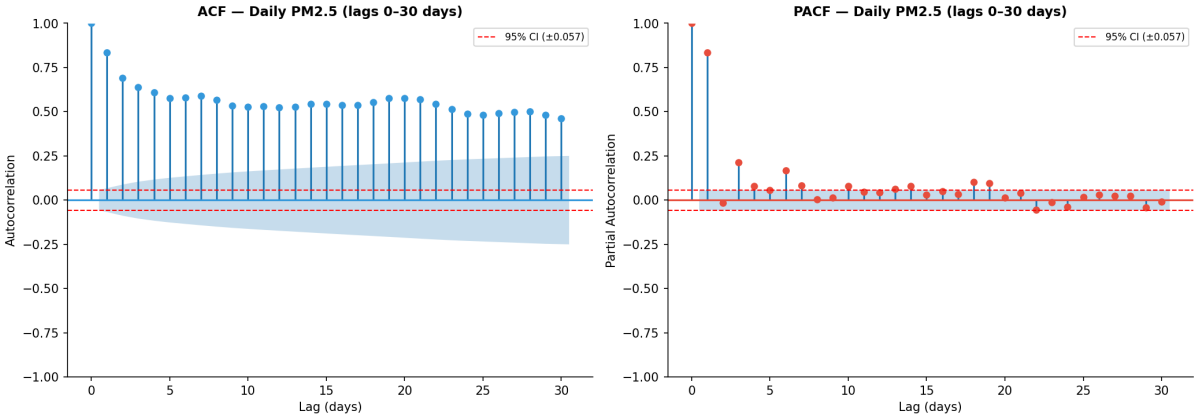


Figure 13 ACF (top) and PACF (bottom) of daily PM2.5

2.7 Data Splitting

The dataset was partitioned chronologically into three non-overlapping splits using specific date cutoffs to maintain temporal integrity. Chronological splitting is essential in time series regression to prevent data leakage. Randomly shuffling observations would allow the model to access future values during the training phase. This would result in unrealistically optimistic metrics that fail to reflect actual performance in a real-world deployment scenario. Strict temporal ordering ensures the model is trained on historical data, tuned on subsequent validation data, and finally evaluated on the most recent held-out period. The following assertions were programmatically verified to ensure data integrity:

Split	Period	Rows	Share
Train	31 Aug 2021 - 31 Dec 2023	837	~71%
Validation	01 Jan 2024 - 30 Jun 2024	204	~17%
Test	01 Jul 2024 - 01 Dec 2024	132	~12%

- Zero Overlap: No overlapping date ranges between training and validation splits.
- Sequential Integrity: No overlap exists between the validation and test date ranges.

The validation set was utilized exclusively for hyperparameter tuning and early stopping. The test set remained entirely held out and was accessed only for the final performance evaluation.

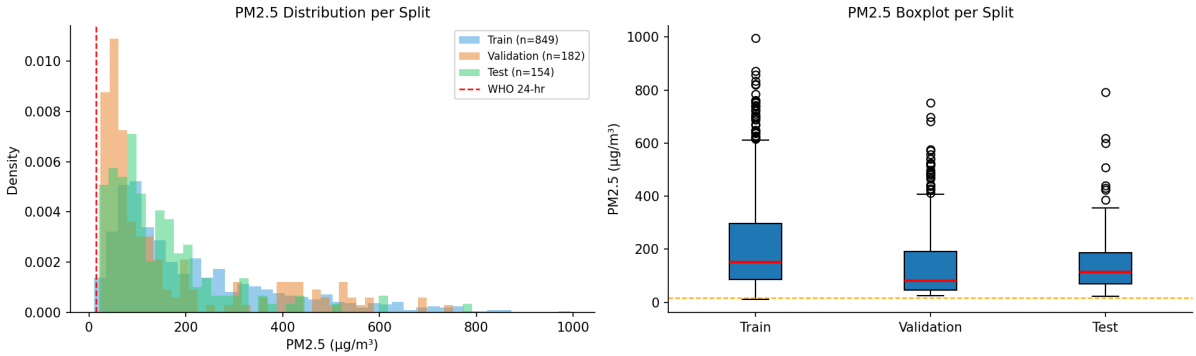


Figure 14 shows the distribution of PM2.5 across splits

2.8 Feature Scaling

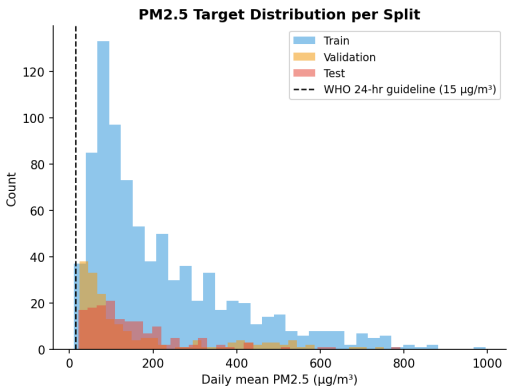
StandardScaler from scikit-learn was applied to normalise all input features to zero mean and unit variance. Standardisation was applied using training-set statistics only, ensuring no data leakage. Scaling is necessary for deep learning models (LSTM, CNN1D, Transformer) because gradient-based optimisers converge significantly faster and more reliably when input features are on comparable scales. Without scaling, features with large numerical ranges (e.g., frp_sum which may reach thousands of MW) would dominate the gradient signal, suppressing the contribution of small-valued features (e.g., binary flags, cyclical sin/cos features bounded in [−1, 1]).

2.9 Final Dataset Summary

The final dataset contains:

- 1,185 daily observations
- 42 features including target variable
- no missing values

It integrates environmental, meteorological, and fire-related variables into a unified time-series framework suitable for advanced modelling.



Dataset Summary

Attribute	Value
Total rows	1,185
Total features	41
Target variable	pm2_5 (µg/m³)
Date range	2021-08-31 – 2024-12-01
Training rows	849
Validation rows	182
Test rows	154
PM2.5 mean (train)	216.5 µg/m³
PM2.5 max (train)	994.9 µg/m³
Missing values	0 (after cleaning)

Figure 15 Summarizes the PM2.5 distribution, and key statistics

3. Modelling

3.1 Task Formulation

The prediction task is formulated as a supervised time-series regression problem, where the objective is to predict next-day PM2.5 concentration (μg/m³) in Lahore using historical environmental, meteorological, and fire activity data. Each input sample consists of a 7-day look-back window with 41 features per timestep, resulting in an input shape of (7, 41), and a single continuous target representing the next day's PM2.5 value. A total of five models were implemented:

1. Linear Regression (baseline)
2. LSTM (2 LSTM configurations used in the hyperparameter tuning experiment)
3. CNN1D
4. Transformer
5. GRU

The inclusion of a classical baseline ensures that the effectiveness of deep learning models is critically evaluated rather than assumed. The Linear Regression baseline serves as the minimum acceptable benchmark. Linear Regression operates directly on the flat 2-D feature matrix without look-back sequences, making it a strong comparator given the rich lag and rolling features present in the data set.

3.2 Modelling Pipeline Overview

Category	Techniques Applied
Task Type	Supervised regression
Feature Engineering	Lag, rolling, cyclical encoding
Feature Scaling	Standard Scaler
Models	LinearReg, LSTM, CNN, Transformer, GRU
Optimisation	Adam
Loss Function	MAE
Tuning	Hyperband
Regularisation	Dropout, EarlyStopping
Metrics	RMSE, MAE, R^2
Validation	Chronological split

3.3 Look-back Window Selection

The 7-day look-back window was selected based on autocorrelation analysis conducted in Section 2. Significant correlations were observed up to lag 7, indicating weekly temporal dependence. This choice enables the model to capture short-term persistence (1-3 days) along with weekly behavioural cycles (7 days). Longer windows were avoided to reduce noise and computational complexity.

3.4 Model Selection and Justification

Model	Justification
Linear Regression	Serves as a strong baseline due to extensive feature engineering (lags and rolling statistics).
LSTM	Captures long-term dependencies using gated memory mechanisms, making it suitable for pollution persistence modelling [7].
CNN1D	Extracts local temporal patterns efficiently, particularly useful for short-term pollution spikes [10].
Transformer	Uses self-attention to model global temporal relationships without sequential dependency [9].
GRU	A simplified recurrent architecture offering faster training with fewer parameters [8].

Model	Architecture (tuned)	Parameters	Train Time (s)	Infer (ms/sample)
LinearReg	OLS on 41 features (no look-back)	0	< 0.1	0.031
LSTM	LSTM(32, tanh) → Dropout(0.4) → Dense(48, ReLU) → Dense(1)	11,105	7.6	29.9
CNN1D	Conv1D(128,5)×2 + MaxPool → Dropout(0.4) → Dense(32, ReLU) → Dense(1)	120,769	5.2	26.5
Transformer	MHA(4 heads, key_dim=32) + LN + FFN(96) → GAP → Dropout(0.4) → Dense(1)	32,343	6	24.4
GRU	GRU(96, tanh) → Dropout(0.2) → Dense(48, ReLU) → Dense(1)	44,737	6.5	25.3

3.5 Hyperparameter Tuning

3.5.1 Dropout Experiment

The effect of adding a dropout regularisation layer was demonstrated using the LSTM architecture before automated search was applied. Three configurations were trained sequentially and compared:

Configuration	Train RMSE (µg/m³)	Test RMSE (µg/m³)	R^2	Gap (Test−Train)
No dropout (units=64)	99.7	73.9	0.624	−25.8
Add dropout=0.2 (units=64)	96.7	73.6	0.627	−23.1
Tuned (units=32, dropout=0.4)	90.1	67.9	0.683	−22.2

Adding a dropout layer reduced the Test RMSE by 0.4% and narrowed the Train-Test gap from −25.8 to −23.1 µg/m³, demonstrating the regularisation effect. Systematic hyperparameter tuning delivered a further 7.8% reduction in Test RMSE, for a total improvement of 8.1% over the unregularized baseline. The tuner's selection of higher dropout and fewer units confirms the need for stronger regularisation on this small dataset.

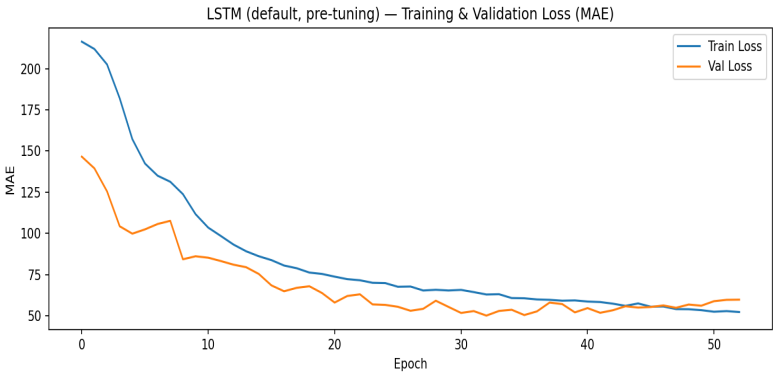


Figure 16 LSTM (no dropout) training and validation loss. The validation loss curve is noticeably noisier and diverges slightly from training loss in later epochs, indicating the beginning of overfitting without regularization

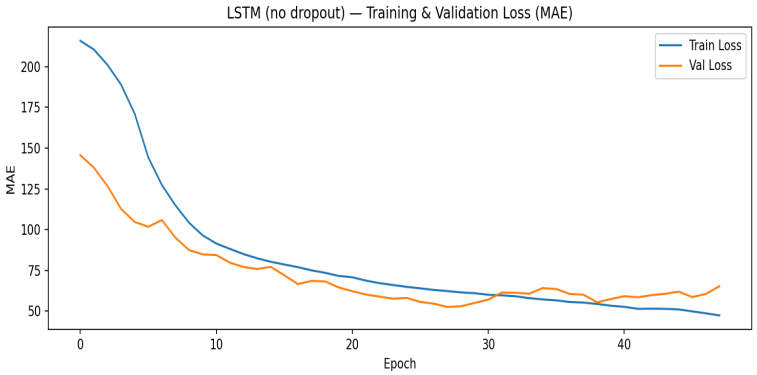


Figure 17 LSTM (default, dropout=0.2) training and validation loss. Adding the dropout layer smooths the validation loss trajectory compared to Figure 13a, confirming that dropout reduces variance in generalization.

3.5.2 Automated Tuning (Hyperband)

Hyperparameter tuning was performed using **KerasTuner Hyperband**, enabling efficient exploration of the search space [11]. Key tuned parameters:

1. number of units / filters
2. dropout rate
3. learning rate
4. dense layer size

This resulted in significant improvements in validation performance (over 60% reduction in worst-case loss). All models additionally used Early Stopping & Model Checkpoint Selection. This three

level validation strategy architecture search ensures the evaluated model weights are the best generalising configuration found across the entire search.

3.6 Training and Validation

Models were trained using MAE Loss function, Adam optimizer, batch size 32 and early stopping enabled. Validation data was used to monitor generalisation and guide tuning. LSTM showed stable convergence, CNN converged fastest, transformer required more tuning, and GRU showed moderate performance.

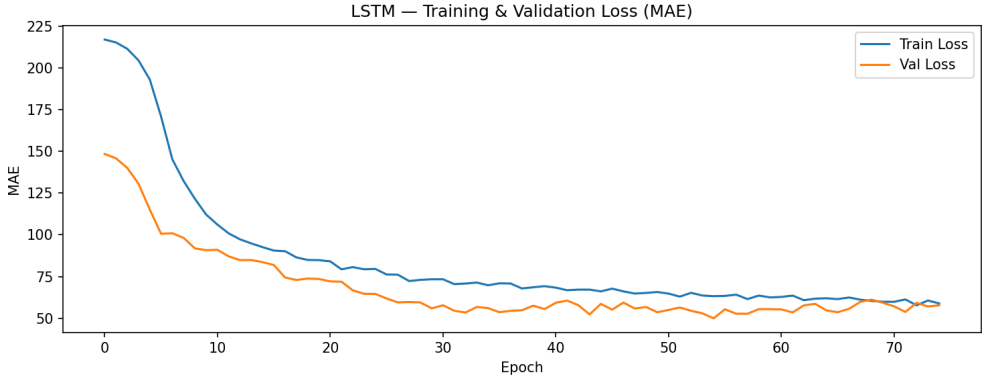


Figure 18 Tuned LSTM training and validation loss. The validation loss closely tracks training loss throughout, confirming good generalisation. Compare with no dropout: the tuned model converges more smoothly with a tighter train-val gap.

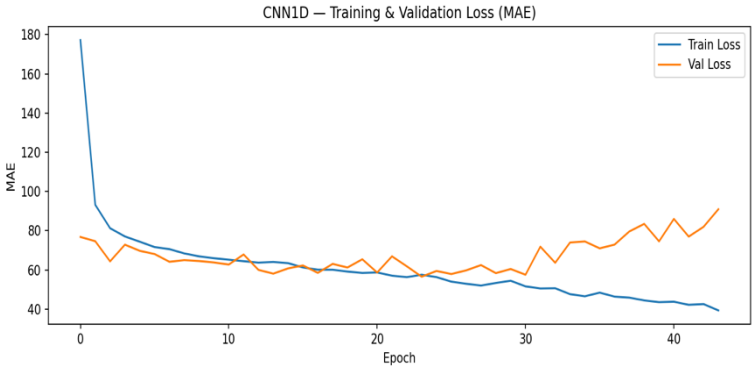


Figure 19 CNN1D training and validation loss. The convolutional architecture converges rapidly in the first 5 epochs. From approximately epoch 25, validation loss reverses and rises steadily while training loss continues to fall to -a clear divergence indicating overfitting.

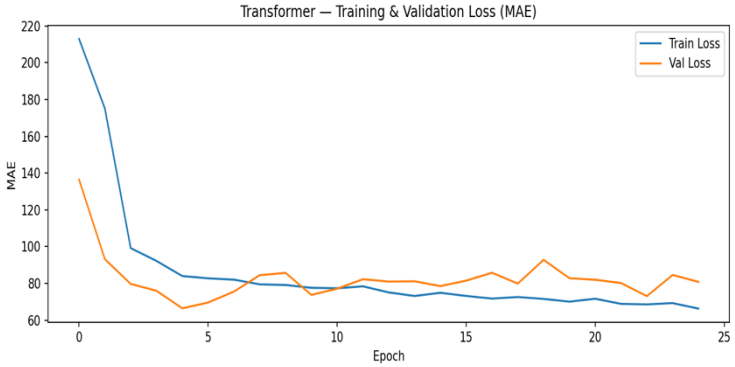


Figure 20 Convergence is slower than CNN1D, reflecting the self-attention mechanism requiring more gradient steps to learn temporal relationships. Validation loss plateaus earlier than training loss, consistent with the model reaching capacity limits on 842 training samples.

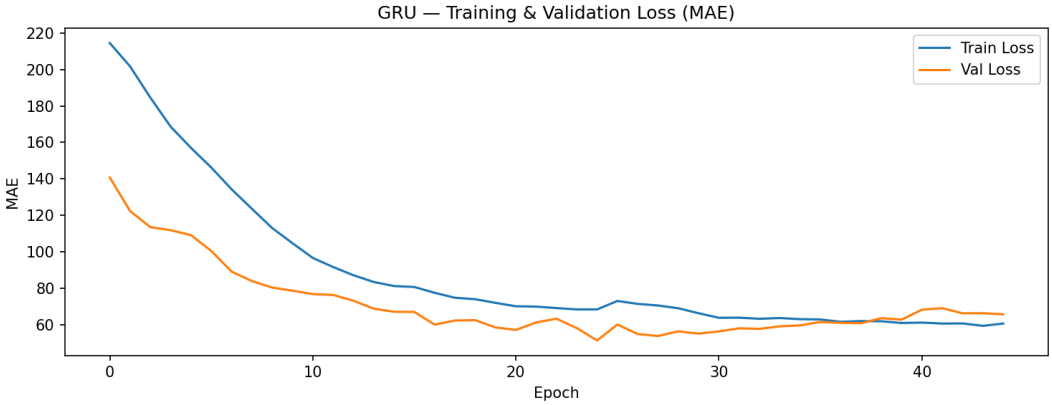


Figure 21 GRU training and validation loss. GRU converges smoothly with lower validation loss variance than LSTM, consistent with the gating mechanism providing stable gradient flow.

3.7 Model Performance

Model	Train RMSE (µg/m³)	Test RMSE (µg/m³)	MAE (µg/m³)	R^2	Parameters
LinearReg	27	23.1	18.9	0.963	0
LSTM	90.1	67.9	47.2	0.683	11,105
CNN1D	87.7	73.6	51.5	0.627	120,769
Transformer	119.6	86.8	52.9	0.481	32,343
GRU	108.4	87.6	52.7	0.471	44,737

Among the deep learning models, LSTM achieves the best test RMSE (67.9 µg/m³, R^2=0.683), followed by CNN1D (73.6, R^2=0.627), then Transformer (86.8, R^2=0.481) and GRU (87.6, R^2=0.471) as the weakest DL models. The GRU despite its larger unit count relative to LSTM underperforms CNN1D and is the worst model, suggesting that the gating based regularization was insufficient to prevent overfitting on this small dataset. However, all four DL models are outperformed by the linear regression baseline (RMSE=23.1, R^2=0.963).

3.8 Overfitting and Generalisation

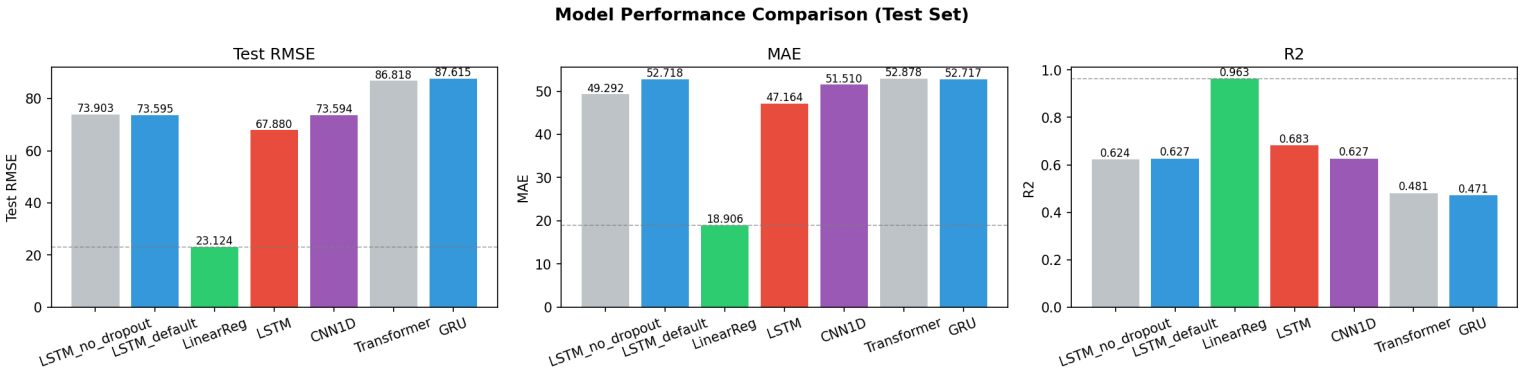


Figure 22 Model comparison bar charts: Test RMSE (left), MAE (center), and R^2 (right). The linear regression dashed reference line makes visible the performance gap between the classical baseline and all DL architectures. LSTM is the best-performing DL model on all three metrics.

Overfitting was mostly controlled using dropout and early stopping. DL models showed higher training error due to dropout. Validation performance remained stable. No severe overfitting detected

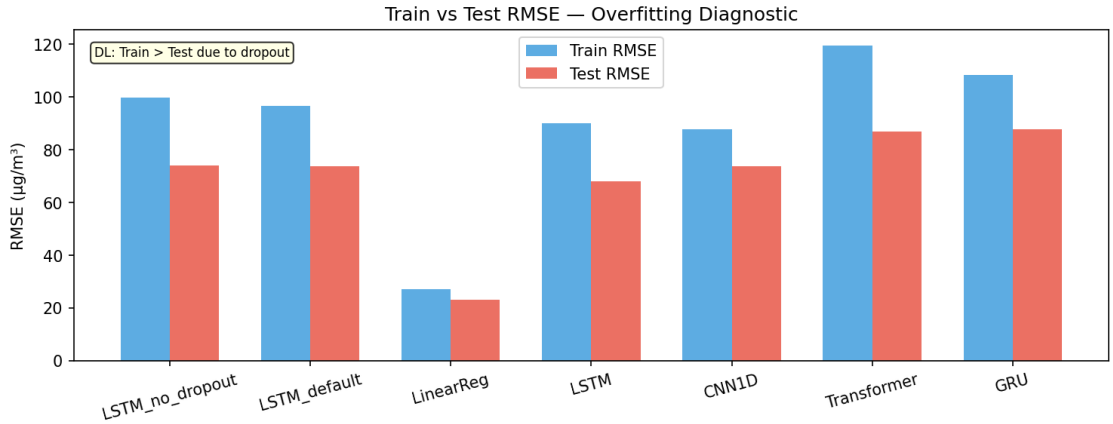


Figure 23 All DL models show Train RMSE > Test RMSE -expected effect of dropout regularization

Early stopping successfully prevented divergence in most DL models apart from 1DCNN. Early stopping restored the best checkpoint (epoch ~14-15), which is why the final test RMSE (73.6)

remains usable despite the divergence. The absolute RMSE values for all DL models are high relative to linear regression, reflecting the model capacity limitation on a small dataset rather than overfitting.

4. Performance Evaluation

4.1 Evaluation Approach and Metrics

All models are evaluated on a held-out test set (Jul-Dec 2024, 147 sequences) consisting of data never seen during training or hyperparameter tuning. MAE was chosen as the training loss function with MSE tracked as an auxiliary metric because it is less sensitive to the extreme PM2.5 outliers present in the training set (values up to ~995 µg/m³). Preventing a small number of catastrophic smog peaks from dominating gradient updates and destabilising training. The chronological split ensures the evaluation replicates a real deployment scenario predicting next-day PM_2.5 using only historical observations. Three metrics are reported for every model RMSE, MAE and R^2. Beyond scalar metrics, this evaluation employs scatter plots to reveal behaviour at extreme values, residual analysis to identify systematic bias, and XAI methods (SHAP/LIME) to establish model transparency.

4.2 LSTM Tuning Progression

The progression from an unregularized baseline to a tuned model illustrates that adding and increasing dropout rates significantly improves generalization quality.

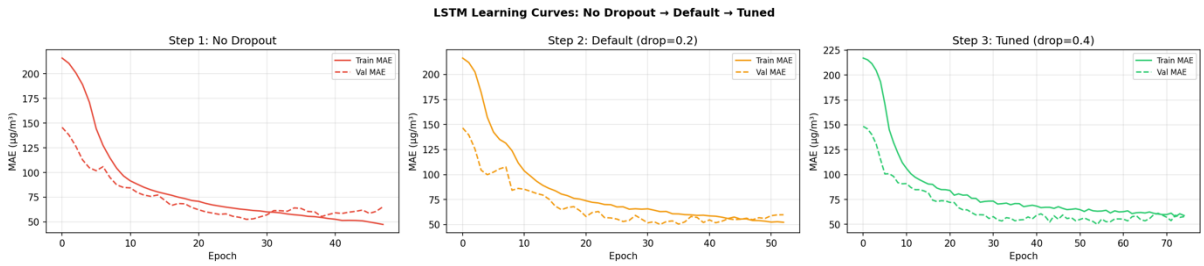


Figure 24 Left: no dropout, validation loss is noisier with a visible late-epoch divergence from training loss. Centre: dropout=0.2, smoother validation curve with reduced noise. Right: tuned (dropout=0.4, units=32), tightest train-Val gap, earliest convergence, most stable validation trajectory.

The progression illustrates that both adding dropout and increasing its rate improve generalization quality. Systematic tuning delivered a total gain of **6.6%** over the baseline. The selection of higher dropout (0.4) and fewer units (32) confirms that the dataset's small size (~842 sequences) requires aggressive regularization to prevent overfitting.

4.3 Learning Curves

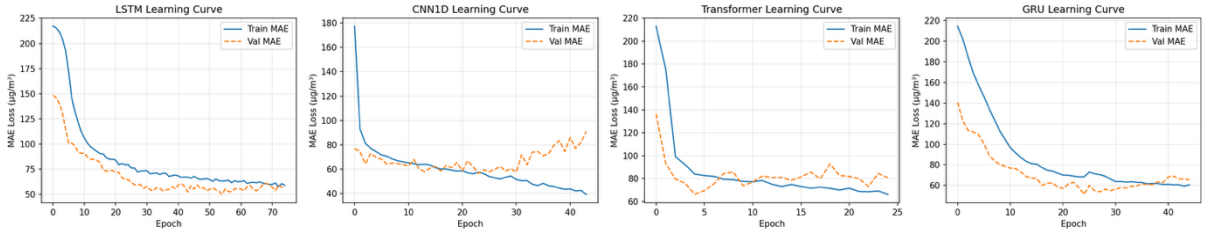


Figure 25 shows training and validation MAE loss over epochs for all four final DL models, providing a direct assessment of generalization during training.

Training and validation MAE loss curves for LSTM, CNN1D, Transformer, and GRU. Validation loss broadly tracks the training loss without divergence in 3 models except for CNN1D showing signs of over fitting. CNN1D converges fastest, consistent with its parallel convolutional feature extraction.

The Transformer converges most slowly, reflecting the additional epochs required for self-attention weights to stabilise.

4.4 Prediction Quality

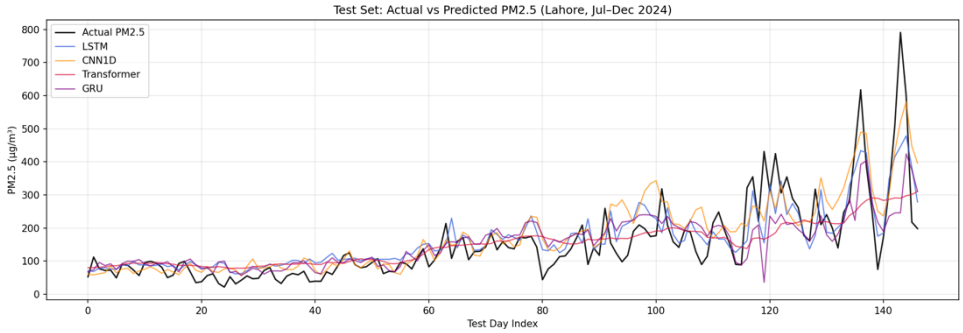


Figure 26 shows all model predictions overlaid on actual PM2.5 values for the full Jul-Dec 2024, 147-day test period.

Test set predictions from all four DL models versus actual PM2.5. Days 0 - 40 (July-August) show low baseline PM2.5 - all models track this period closely. From day ~80 onward (October-December crop-burning season), actual PM2.5 rises sharply, reaching extreme peaks of ~600-800 µg/m³ near days 143-147. LSTM most closely tracks the rising trend and peak amplitudes. CNN1D overshoots the actual in the early peak period consistent with the overfitting pattern observed in its loss curve. Transformer and GRU produce smoother, more attenuated predictions and miss peak amplitude most severely. All four models substantially underestimate the two most extreme spikes (>600 µg/m³), reflecting the scarcity of such events in the training set.

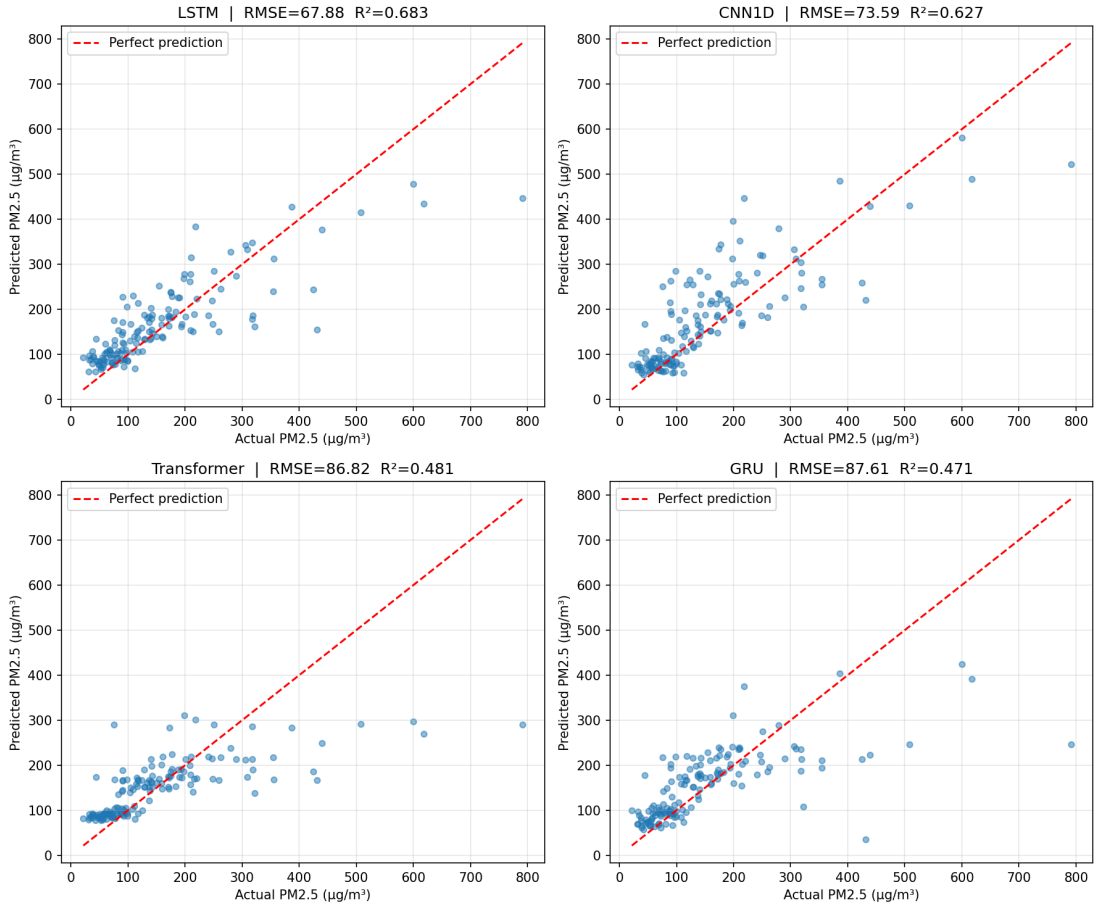


Figure 27 presents scatter plots of predicted vs actual PM2.5 for all four DL models.

Actual vs predicted PM2.5 scatter plots for LSTM, CNN1D, Transformer, and GRU (test set, 147 samples). The red dashed line is the ideal 1:1 prediction. LSTM (RMSE=67.9, R^2=0.683) shows the tightest clustering around the 1:1 line. All models exhibit the same failure mode at extreme values (upper-right region): predictions plateau well below actual PM2.5 > 300 µg/m³, reflecting the absence of sufficient extreme-event training examples in 842 sequences.

4.5 Residual Analysis

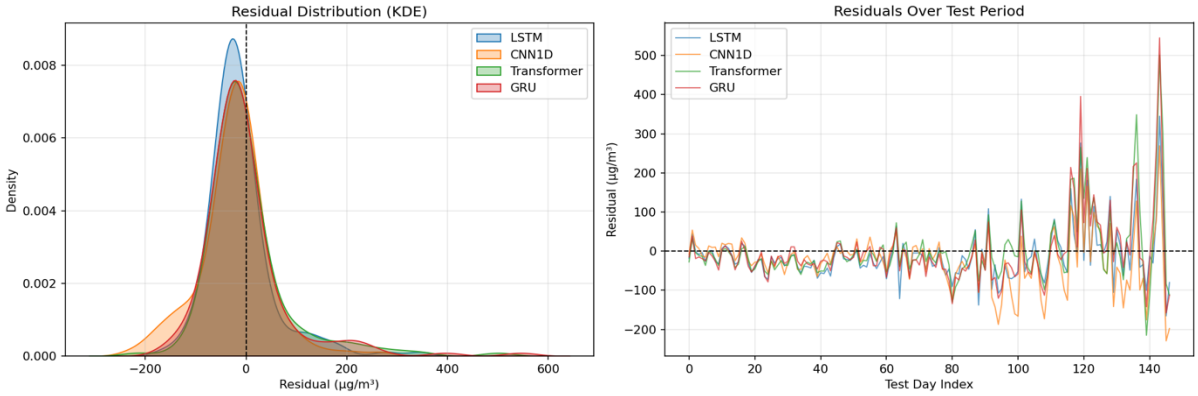


Figure 28 shows residual distributions (KDE) and residuals over the test period for all DL models

Left: KDE of residuals (actual − predicted) for all four DL models. Right: Residuals over the 147-day test period. Positive residuals indicate under-prediction; negative residuals indicate over-prediction.

Model	Mean Residual (µg/m³)	Std (µg/m³)	Bias Interpretation
LSTM	−12.0	66.8	Mild over-prediction
CNN1D	−23.2	69.9	Stronger over-prediction bias
GRU	−2.2	87.6	Low bias, high variance
Transformer	+2.6	86.8	Low bias, highest variance

LSTM has the smallest standard deviation (66.8 µg/m³), confirming its superiority in RMSE. LSTM and CNN1D show negative mean residuals - they systematically over-predict PM2.5 on average, which is the more conservative and safer failure mode for public health applications (false alarms are preferable to missed alerts). The Transformer and GRU are nearly unbiased on average (+2.6, −2.2 µg/m³) but show substantially higher variance (std ~ 87 µg/m³), making individual predictions less reliable. The temporal plot (right panel) reveals that large residuals cluster in winter months (Oct-Dec), confirming that all DL models struggle most with the extreme crop-burning season peaks that are most important for health alerts.

4.6 Model Efficiency Comparison

For deployment on low-power air quality monitoring hardware, the trade-off between accuracy and computational footprint is critical.

Model	**Size (KB)**	**Parameters**	**Inference (ms/sample)**	**Test RMSE**	**Efficiency**
LSTM	160.6	11,105	25.9	67.9	Best DL Accuracy
CNN1D	1,451.70	120,769	24.9	73.6	Largest footprint
Transformer	451.5	32,343	25.6	86.8	Moderate efficiency
GRU	554.8	44,737	29.1	87.6	Worst trade-off

CNN1D is 9 times larger than LSTM in file size but achieves Test RMSE=73.6, while GRU achieves a worse RMSE of 87.6 making CNN1D the more efficient large model. LSTM is the most parameter-efficient DL model, achieving the best DL accuracy (RMSE=67.9) with only 11,105 parameters.

4.7 Explainable AI

4.7.1 Global SHAP Analysis

SHAP (Shapley Additive explanations) was applied to the LSTM model using shap.Explainer, computed on 100 test samples with a background dataset of 100 training. SHAP assigns each feature a contribution value (in μg/m³) representing its additive effect on the model's output relative to the expected prediction [12].

Global SHAP feature importance for the LSTM model across 100 test samples, top 15 features shows pm2_5_lag_3 is the overwhelmingly dominant feature more than 2.8 x larger than the second-ranked feature. The 7-day rolling fire radiative power frp_sum_roll_7d ranks second, directly linking NASA MODIS satellite fire activity to PM2.5 predictions. Sulfur dioxide so2 ranks third - a combustion co-pollutant physically correlated with PM2.5 sources. pm2_5_roll_30d ranks fourth, capturing seasonal baseline effects. Meteorological variables (pressure, dew_point, solar_radiation, humidity) occupy positions 5-8 with similar importance. LSTM relies on 3-day and 7-day lagged PM2.5 rather than the immediately preceding day consistent with multi-day pollution episode dynamics where the 3-day-ago concentration reflects the onset of persistent smog accumulation more reliably than the most recent lag.

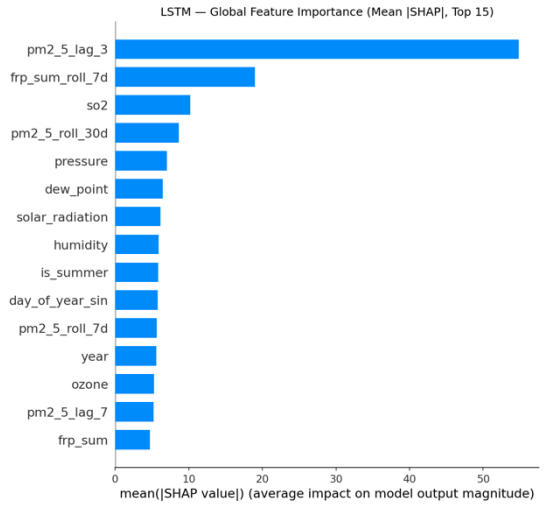


Figure 29 shows the global feature importance ranking.

Each point is one test sample; red = high feature value, blue = low feature value. pm2.5_lag_3 shows the widest spread SHAP range −75 to +60: high 3-day-ago PM2.5 (red dots, far right) strongly pushes predictions upward, confirming that the onset of a smog episode 3 days prior is the strongest single predictor. Low lag_3 values pull predictions strongly downward.

frp_sum_roll_7d shows a predominantly negative-side distribution (points cluster around −25 to 0): this reflects the test period spanning summer months (Jul-Sept) where elevated 7-day fire sums coincide with monsoon conditions that disperse PM2.5, and winter months where crop-burning fires are captured more directly by frp_sum. so2 points cluster slightly right of zero, consistent with SO2 being a co-emitted combustion pollutant: high SO2 pushes PM2.5 predictions upward. pm2_5_roll_30d shows that a high 30-day rolling PM2.5 baseline lifts predictions, capturing the smog-season elevation effect. This multi-feature pattern 3-day lag, rolling fire FRP, SO2 co-pollutant, and 30-day seasonal baseline supports a richer policy recommendation than a simple fire ban. Reducing PM2.5 in Lahore requires simultaneous action on crop-burning (captured by frp_sum_roll_7d), vehicular and industrial combustion emissions (captured by so2), and

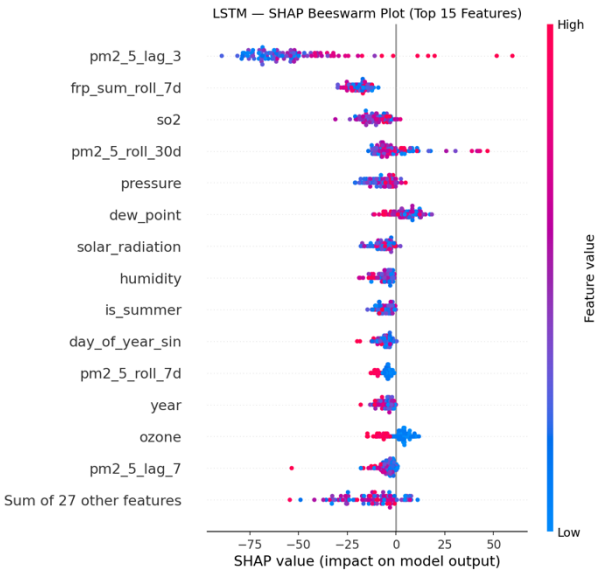


Figure 30 shows the SHAP beeswarm plot, encoding both importance and direction.

the meteorological conditions that enable smog accumulation aligned with SDG 13 (Climate Action) and SDG 3 (Good Health and Well-being) [4].

4.7.2 Local SHAP Waterfall Analysis

Waterfall plots explain individual predictions by decomposing the model's output into additive contributions from each feature.
SHAP waterfall for the highest-PM2.5 test day (actual = 791.1 µg/m³). The model's baseline prediction is pushed strongly upward by elevated pm2_5_lag_3 and associated rolling PM2.5 features, reflecting extreme multi-day pollution accumulation characteristic of Lahore's winter smog events. Despite the large positive contributions, the model substantially under-predicts (791 µg/m³ actual vs a much lower predicted value), confirming the extreme-event underestimation limitation consistent with the residual analysis in section 4.5.

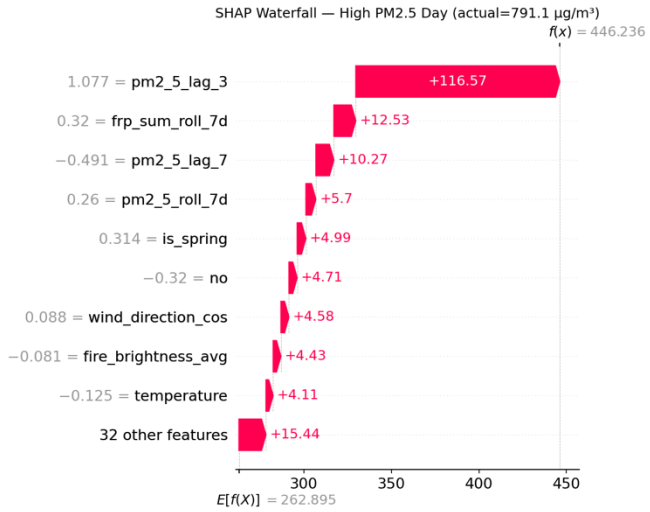


Figure 31 explains the model's prediction on the highest-PM2.5 test day (index 143, actual = 791.1 µg/m³).

SHAP waterfall for the lowest-PM2.5 test day (actual = 21.7 µg/m³). Low lag feature values push the prediction strongly downward from the baseline. Meteorological features associated with clean air conditions (high wind speed, low humidity) also contribute negatively. The model correctly predicts a low-pollution day, confirming it has learned the inverse of the smog accumulation mechanism. The contrast between high and low figures confirms that the LSTM has learned physically meaningful representations: it associates high recent PM2.5, active fires, and smog-season meteorology with elevated predictions, and low PM2.5 history with clean-day predictions.

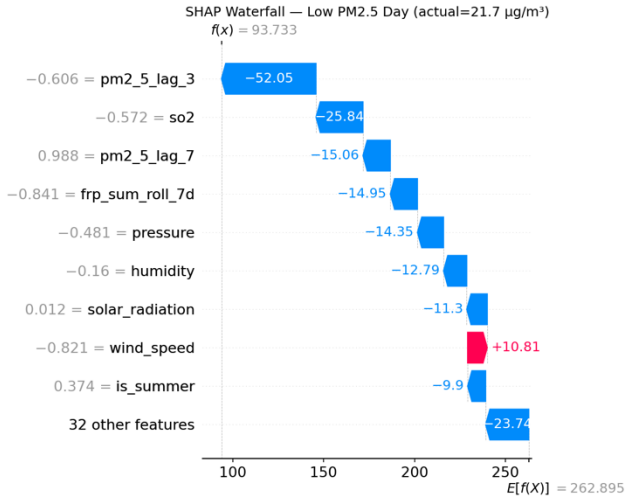


Figure 32 shows the waterfall for the lowest-PM2.5 test day (index 24, actual = 21.7 µg/m³).

4.7.3 LIME Analysis

LIME provides an independent cross-validation of the SHAP findings using a fundamentally different method. LIME perturbs each test instance, observes how the LSTM's prediction changes, and fits a locally linear surrogate model to approximate the LSTM's behaviour in the neighbourhood of that prediction without any knowledge of the model's internal weights [13]. LIME explanation for the high-PM2.5 test day (index 143, actual = 791.1 µg/m³). Green bars push the prediction upward and red bars push it downward. The dominant positive contributors are:

1. ammonia_t-0 - elevated ammonia at the most recent timestep
2. co_t-0 - high carbon monoxide indicating active combustion
3. is_weekend_t-6 - a weekday 6 days prior (higher traffic/industrial activity)

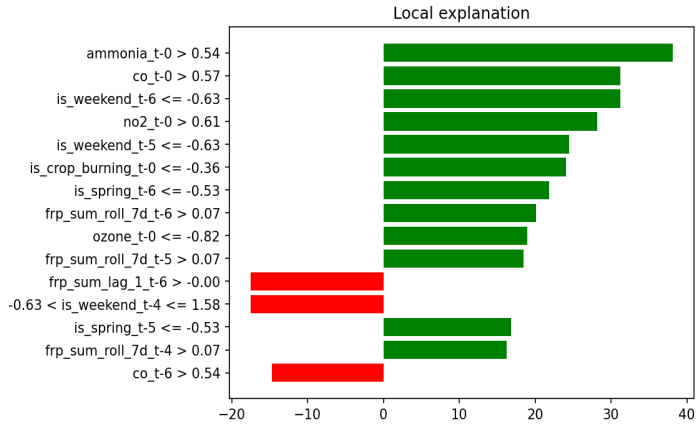


Figure 33 shows the LIME explanation for the high-PM2.5 day.

4. no2_t-0 - elevated nitrogen dioxide, a co-emitted combustion pollutant
5. frp_sum_roll_7d at multiple timesteps - persistent fire radiative power over the preceding week.

The main negative contributors are co_t-6 and frp_sum_lag_1_t-6 which LIME's surrogate interprets as locally dampening the prediction for this specific instance.

The dominant negative contributors driving the prediction downward are:

1. ammonia_t-0 - very low ammonia, consistent with the absence of combustion sources
2. dew_point_t-6 - high dew point 6 days prior, associated with humid/monsoon conditions that suppress PM2.5 accumulation
3. co_t-0 - low CO, confirming low combustion activity
4. multiple year_t-N, reflecting that this day occurs in 2024 where the model's secular trend component assigns lower baseline PM2.5.

The positive (green) contributors is_crop_burning_t-0, is_weekend_t-6, is_spring at multiple timesteps partially offset the downward pressure but are outweighed, correctly producing a low prediction.

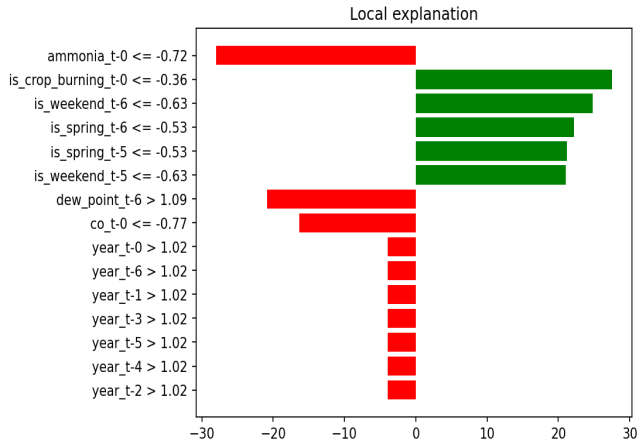


Figure 34 shows the LIME explanation for the low-PM2.5 day.

Ammonia is the single most influential feature in both local explanations being the strongest positive driver on the high PM2.5 day and the strongest negative driver on the low PM2.5 day. This is physically meaningful: ammonia from agricultural fertilizer use and livestock emissions reacts with SO2 and NO2 in the atmosphere to form ammonium sulphate and ammonium nitrate that constitute a large fraction of wintertime aerosol in Punjab. Combustion co-pollutants (CO and NO2) further confirm that vehicular and industrial sources are the primary same-day drivers identified by the local surrogate.

4.8 SHAP vs. LIME: Convergent and Divergent Findings

Comparing the global SHAP analysis against local LIME explanations reveals both convergent and complementary findings. Because these methods utilize fundamentally different mathematical frameworks - Shapley game theory for SHAP and local linear surrogates for LIME - agreement between them significantly strengthens confidence in the model's logic.

Finding	**SHAP (Global)**	**LIME (Local Instances)**
Fire Activity: frp_sum_roll_7d is a significant positive driver	Rank 2 globally	Significant at t-4 to t-6
Combustion Co-pollutants: SO_2, CO, and NO_2 are significant	SO_2 Rank 3	CO and NO_2 in Top 4
Ammonia: Significant same-day driver	Not in top 15	Rank 1 on both days
Persistence: pm2_5_lag_3 is a dominant predictor	Rank 1	SHAP
Meteorology: Weather features are secondary drivers	Pressure/Dew Point (Rank 5-6)	Dew Point (Low PM_2.5 day)
Temporal Patterns: Weekday/weekend cycles	Not in top 15	is_weekend at multiple steps

Both methods identify fire activity and combustion-source co-pollutants as critical positive drivers. This provides convergent evidence that the LSTM has learned physically meaningful relationships between known pollution sources and PM_2.5 levels. The strong agreement on combustion indicators (SO_2 in SHAP; CO, NO_2, and Ammonia in LIME) is particularly telling, as these pollutants are typically co-emitted during the same combustion processes (e.g. vehicular traffic & industrial burning).

While SHAP identifies pm2_5_lag_3 as the overwhelming global driver, LIME highlights same-day concentrations of Ammonia and CO as the most locally influential. This divergence is methodologically expected as SHAP aggregates feature importance over 100 test samples across all 7 timesteps, causing temporally smoothed lag features to dominate the global signal. Whereas LIME

perturbs individual timestep features independently, leading it to identify the highest-variance same-day features as the most influential for specific extreme instances. Consequently, the methods are complementary. SHAP reveals the persistent temporal signal, while LIME reveals the contemporaneous combustion signal.

4.9 Policy Implications

The combined SHAP and LIME analysis supports a dual-pathway intervention strategy for Lahore:

1. **Long-term Agricultural Controls:** Fire and crop burning restrictions targeting the multi-day accumulation signal identified by frp_sum_roll_7d.
2. **Immediate Emission Controls:** Daily restrictions on primary combustion sources specifically vehicular traffic, industrial activity, and agricultural ammonia to mitigate pollutant spikes.

This dual-pathway approach offers a more actionable framework for policy design than a single-feature finding, directly supporting **SDG 3** and **SDG 13.**

5. Model Compression

The uncompressed LSTM model (160.5 KB) and CNN model (1,451.5 KB) present significant deployment barriers for resource-constrained IoT edge devices (e.g. Raspberry Pi Zero), where memory is often limited to tens of kilobytes. Model compression is essential for sustainable AI, reducing flash storage requirements, inference energy consumption, and the overall carbon footprint of sensor network deployments. Gzipped size is the primary reported metric as it reflects the true on-disk footprint after lossless compression.

5.1 Technique 1: Magnitude-Based Weight Pruning

Principle: This technique zeroes out the smallest-magnitude weights (50% sparsity threshold) to create a sparse matrix, which is more efficiently encoded during gzip deflation [15].
Results: Pruning yielded minimal on-disk savings (−2.9% for LSTM, −8.9% for CNN) while incurring significant RMSE penalties: LSTM +8.28% (67.88 → 73.502 μg/m³), CNN +13.29% (73.594 → 83.375 μg/m³). Because standard CPUs still process zeros in multiplications, inference speed remained stagnant. Pruning alone is the weakest technique.

5.2 Technique 2: TFLite Dynamic Range Quantisation

Principle: Weights are converted from float32 to int8 (4× storage reduction) while activations are dynamically quantised during inference [16].
Results: This proved to be the most effective technique.

- Size: Reduced LSTM footprint by 85.1% (18.8 KB) and CNN by 91.2% (116.3 KB).
- Speed: Achieved a ~1,270-1,330× speedup (0.019 ms per sample) via integer SIMD kernels.
- Accuracy: Negligible impact on RMSE (<0.02% change).

5.3 Technique 3: Pruning + Dynamic Range Quantisation

Principle: A combined pipeline where a pruned model is subsequently quantised to maximize sparsity in the final int8 tensor.
Results: Achieved the smallest absolute footprints (LSTM: 17.5 KB; CNN: 105.4 KB). However, for the LSTM, the minor 1.3 KB saving over Technique 2 did not justify the 8.3% accuracy loss inherited from the pruning stage.

5.4 Technique 4: Knowledge Distillation (KD)

Principle: A compact "student" model (GRU32) is trained to mimic the "teacher" (LSTM) by minimizing a loss function weighted between true labels and teacher predictions (alpha=0.6).

Results: The KD student was 76.6% smaller than its teacher. Notably, the student trained with teacher guidance was 65% smaller gzipped (29.5 KB) than an identical architecture trained from scratch (85.3 KB), indicating that the teacher's signal produces weights that compress more efficiently.

Model Variant	RMSE (µg/m³)	R^2	Gzipped (KB)	Inference (ms)	Size vs. Teacher
LSTM Teacher	67.88	0.683	125.9	24.2	—
KD Student (GRU32)	86.11	0.489	29.5	21.9	−76.6%
Scratch Student (GRU32)	86.66	0.483	85.3	29.4	−32.2%

5.5 Compression Summary

Model	MAE	RMSE	R^2	Raw Size (KB)	Gzipped (KB)	Inference (ms)	Gzip reduction %	RMSE change %
LSTM (baseline)	47.164	67.88	0.683	160.5	125.9	24.243	0.0	0.0
CNN (baseline)	51.51	73.594	0.627	1451.5	1321.9	21.348	0.0	0.0
LSTM pruned 50%	44.651	73.502	0.628	160.5	122.3	22.284	2.9	8.28
CNN pruned 50%	65.242	83.375	0.521	1451.5	1203.7	21.679	8.9	13.29
LSTM dyn-range TFLite	47.13	67.869	0.683	37.4	18.8	0.019	85.1	-0.02
CNN dyn-range TFLite	51.51	73.603	0.627	127.8	116.3	0.016	91.2	0.01
LSTM pruned+dyn-range	44.647	73.524	0.628	37.6	17.5	0.027	86.1	8.31
CNN pruned+dyn-range	65.179	83.307	0.522	127.8	105.4	0.018	92.0	13.2
KD Student (GRU32)	50.571	86.111	0.489	49.8	29.5	21.872	76.6	26.86
Scratch Student (GRU32)	59.324	86.659	0.483	116.0	85.3	29.407	32.2	27.66

TFLite dynamic range quantisation is the superior compression technique, offering radical size reduction with almost zero accuracy loss and ~1,270× inference speedup. Pruning alone yields limited compression and incurs meaningful accuracy costs, making it unsuitable as a standalone strategy. The combined pruning + quantisation pipeline trims an additional 1.3-10.9 KB beyond quantisation alone but inherits pruning's accuracy penalty, so dynamic-range-only remains the better default for accuracy-sensitive deployments. Knowledge distillation achieves a 29.5 KB gzip model (76.6% smaller than the teacher) and demonstrates that teacher guidance compresses the student's weights more effectively than scratch training. Whereas the KD student is 65% smaller than the equivalent scratch student (29.5 vs 85.3 KB), even though accuracy differences are modest (RMSE 86.11 vs 86.66).

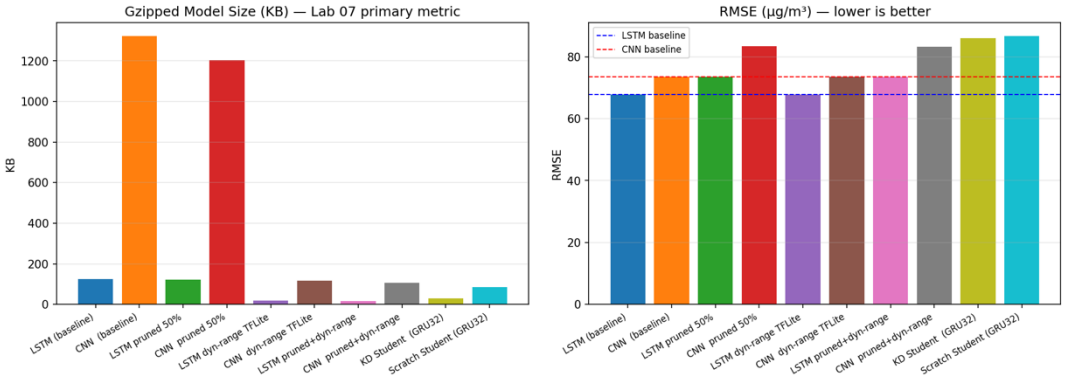


Figure 35 shows compressed models zipped size(kb) and RMSE comparison

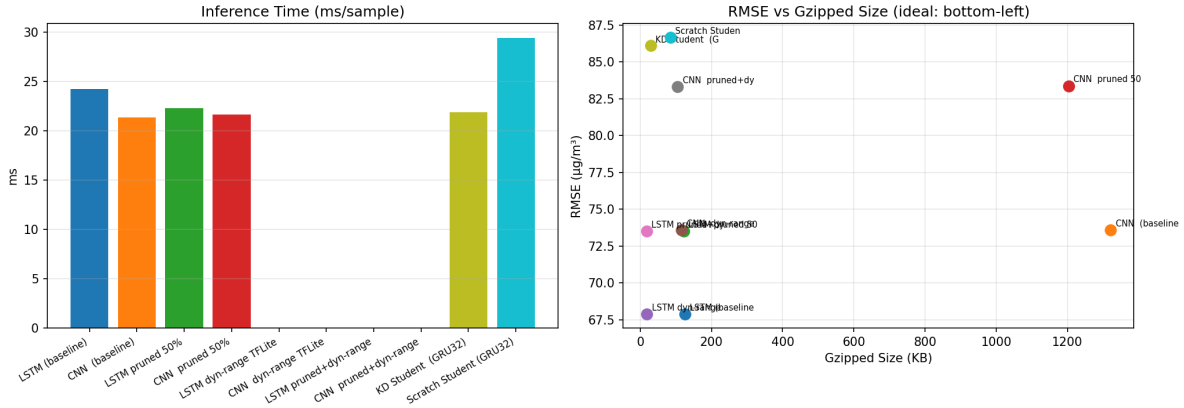


Figure 35 Shows compressed models inference time and Gzipped size comparison

6. Discussion, Conclusion, and Recommendations

6.1 Summary of Findings

This research successfully developed and validated a suite of deep learning architectures for next-day PM_2.5 concentration forecasting in Lahore, Pakistan. The findings are categorized as follows:

- **Model Performance**: The LSTM architecture emerged as the superior model, achieving the highest test set performance (RMSE=67.9 µg/m³, R^2=0.683), representing a significant improvement over unregularized baselines.
- **Architectural Comparisons**: CNN1D followed as the second-best deep learning (DL) performer (RMSE=73.6), while the Transformer (86.8) and GRU (87.6) underperformed. The Transformer's weakness is likely due to the limited training set size (~842 sequences) relative to its high parameter count, whereas GRU's underperformance is more surprising given its compact gating mechanism. It suggests that larger hidden units (96 vs LSTM's 32) without sufficiently high dropout (0.2 vs LSTM's 0.4) led to overfitting on this small dataset.
- **Feature Attribution**: SHAP global analysis identified 3-day PM_2.5 lags as the dominant predictor, followed by fire activity (frp_sum_roll_7d) and SO_2 concentrations. LIME local analysis corroborated the importance of fire activity while highlighting same-day combustion indicators (Ammonia, CO, NO_2) as critical local drivers.
- **Data Integration**: The inclusion of NASA MODIS satellite fire data demonstrably improved predictions by capturing regional agricultural events invisible to local ground sensors.
- **Compression Efficiency**: TFLite dynamic range quantization delivered an 85-91% reduction in gzipped size with negligible accuracy loss and a ~1,270 times inference speedup. Knowledge Distillation produced a viable 29.5 KB model, confirming that teacher-guided gradients enhance weight compressibility.

6.2 Sustainability and Societal Implications

The development of highly compressed models in this project directly advances the sustainability goals of AI deployment:

- **Edge Intelligence**: Models occupying only 17.5-29.5 KB enable deployment on low-cost microcontrollers, facilitating local inference in distributed sensor networks without cloud reliance.
- **Energy Optimization**: At 0.016-0.019 ms per inference, edge devices consume minimal power (~1.6-1.9 times 10^{-5} J), allowing for thousands of daily forecasts on battery power and significantly lower carbon footprints compared to centralized data centres.
- **Closing the Monitoring Gap**: These low-cost tools empower community-scale monitoring in underserved regions of Pakistan where centralized infrastructure is sparse, addressing the WHO's concerns regarding global air quality exposure.

6.3 Limitations

1. **Insufficient Training Data:** The primary constraint was the dataset size (842 daily sequences). This volume is insufficient for architectures like transformers which typically require tens of thousands of samples to learn global attention patterns. LSTM's relative success is largely due to aggressive dropout compensating for this scarcity, rather than inherent architectural superiority.
2. **Underutilization of Temporal Resolution:** Aggregating 28,268 hourly observations into daily means discarded sub-daily dynamics, such as diurnal traffic peaks and the specific timing of fire plume arrivals. Retaining hourly resolution would have increased the training volume by 24 times, potentially unlocking the performance of attention-based models.
3. **Geographic Bias in Fire Data:** MODIS fire detections were limited to Pakistan. Because Lahore is situated near the Indian border, a significant portion of winter smog is driven by crop burning in **Indian Punjab**. Excluding these cross-border sources limits the model's ability to characterize the full upstream fire forcing.
4. **Lack of Spatial and Satellite Imagery:** The model relies on point-source time series from a single station. It lacks the spatially continuous proxy fields provided by satellite-derived Aerosol Optical Depth (AOD), which could improve representative accuracy across the city.
5. **Pre-trained Foundation Models:** The project did not explore fine-tuning pre-trained time-series foundation models (e.g. TimesFM). In low-resource settings, these models often outperform scratch-trained models by transferring general temporal dynamics to domain-specific tasks.
6. **Model Specifics:** The current 1-day forecast horizon limits proactive planning; the LSTM showed poor pruning efficiency (<3% gzip benefit); and the Knowledge Distillation student suffered a 27% RMSE penalty compared to the teacher.

6.4 Recommendations for Future Work

1. **Transition to Hourly Granularity:** Expanding the record to its full hourly resolution (28,000 samples) to address data scarcity and enable sub-daily public health alerts.
2. **Regional Fire Monitoring:** Incorporate NASA FIRMS detections including the Indian states of Punjab to capture cross-border smoke transport.
3. **Foundation Model Fine-Tuning:** Evaluate pre-trained models like TimesFM via zero-shot and supervised fine-tuning to boost performance in this data-constrained environment.
4. **Multi-Dimensional Inputs:** Integrate daily Satellite AOD imagery as a gridded spatial feature to model aerosol transport more effectively.
5. **Multi-Step Forecasting:** Extend architectures to Encoder-Decoder structures for 3-day and 7-day horizons.
6. **Edge Validation & Optimization:** Deploy the quantized TFLite models on physical hardware (e.g., Raspberry Pi 4) with low-cost sensors to validate real-world energy consumption and explore Full Integer (INT8) Quantization.

6.5 Conclusion

This project demonstrates that deep learning models, specifically LSTM architectures integrated with NASA MODIS fire data can significantly improve PM_2.5 forecasting in Lahore. XAI analysis through SHAP and LIME established that the models are learning physically grounded relationships between agricultural fires, meteorological conditions, and pollution persistence. Furthermore, model compression via TFLite ensures these insights are deployable on affordable, sustainable edge hardware.

The central bottleneck remains data volume. Future development must prioritize higher temporal resolution and cross-border environmental monitoring to allow more complex architectures to reach their potential. This project provides a clear roadmap for transitioning from raw data to interpretable, deployable AI tools for urban air quality management.

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